2 **(a)** **(i)** State what is represented by a gravitational field line.

.....

.....

..... [2]

(ii) The Earth may be considered as a uniform sphere, as shown in Fig. 2.1.

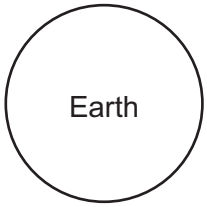


Fig. 2.1

On Fig. 2.1, draw field lines to represent the Earth's gravitational field outside the Earth. [2]

(b) The Earth's magnetic field may be considered as being due to the Earth acting as a long solenoid, as shown in Fig. 2.2.

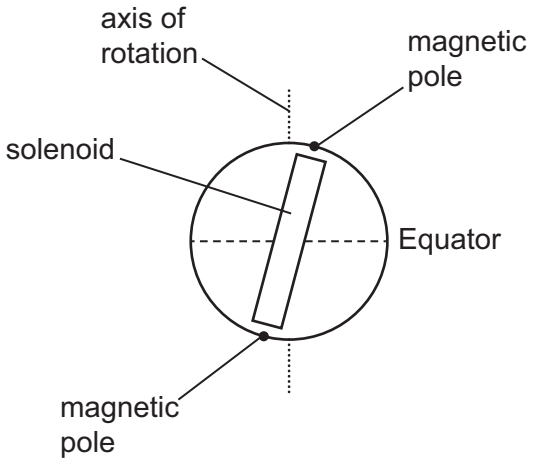


Fig. 2.2

The magnetic poles do not align with the geographic poles, which are on the axis of rotation.

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

Fig. 2.3 is a copy of Fig. 2.2 without the labels but with two magnetic field lines shown.

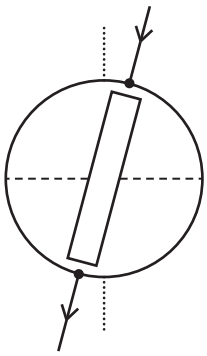


Fig. 2.3

(i) On Fig. 2.3, label the magnetic poles with the letters N and S to indicate which one is the magnetic N pole and which one is the magnetic S pole. [1]

(ii) On Fig. 2.3, draw field lines to represent the Earth's magnetic field outside the Earth. [2]

(c) An observer moves around the surface of the Earth.

(i) Use your answer in **(a)(ii)** to explain why the observed gravitational field of the Earth does not vary around the surface.

.....

.....

..... [2]

(ii) With reference to your answer in **(b)(ii)**, describe how the observed magnetic field of the Earth varies around the surface.

.....

.....

.....

.....

..... [3]

[Total: 12]

DO NOT WRITE IN THIS MARGIN